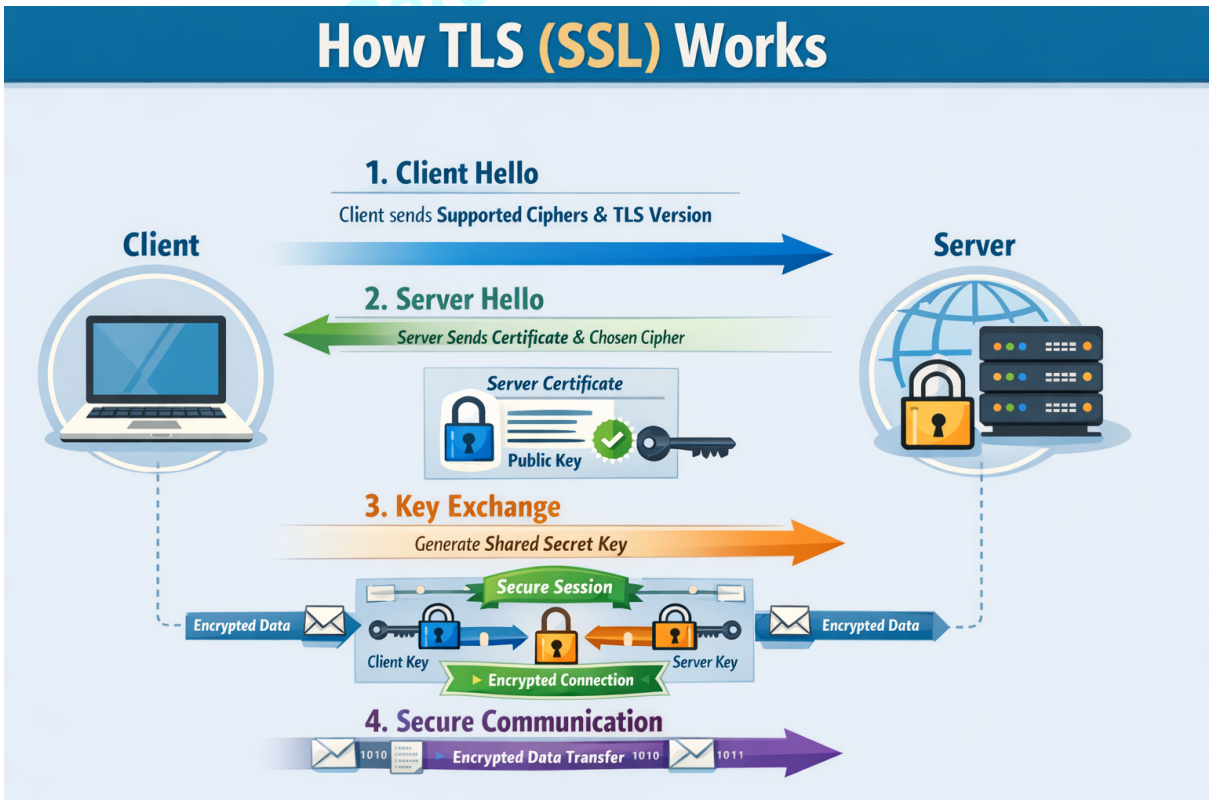
What is an SSL Certificate?

An SSL (Secure Sockets Layer) certificate encrypts data between a web server and a browser, ensuring secure communication. DigiCert is a trusted Certificate Authority (CA) that issues these certificates.





TLS 1.2 vs TLS 1.3 handshake differs:

- **TLS 1.3 is faster (fewer round trips)** and removes older/insecure options.
- **TLS 1.3 does not use RSA key exchange**; it uses **(EC)DHE** (Diffie–Hellman) so it provides **forward secrecy by default**.
- In **TLS 1.3**, more of the handshake is encrypted earlier, and the message list changed (e.g., "ChangeCipherSpec" is basically legacy).

Installing a DigiCert SSL Certificate on Nginx

🛠️ Steps to Obtain and Install a DigiCert SSL Certificate on Nginx

1. Generate a Certificate Signing Request (CSR) Using OpenSSL

On your server, generate a private key and CSR:

```
bash

openssl genrsa -out /etc/ssl/private/your_domain_name.key 2048
openssl req -new -key /etc/ssl/private/your_domain_name.key -out
/etc/ssl/private/your_domain_name.csr
```

Fill in the required information, especially the **Common Name**, which should be your domain name (e.g., `www.yourdomain.com`).

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2. Submit the CSR to DigiCert

- Log in to your DigiCert account.
- Navigate to the SSL certificate section and select "Rekey" or "New Order" (depending on your situation).
- Paste the CSR into the provided field.
- Complete the domain validation process as per DigiCert's instructions.

3. Download the Issued Certificate

Once DigiCert validates your domain, they will issue your SSL certificate. You can download it from your DigiCert account. The certificate files typically include:

- **Primary Certificate**: `your_domain_name.crt`
- **Intermediate Certificate(s)**: `DigiCertCA.crt`, `DigiCertHighAssuranceCA-3.crt`, etc.

4. Prepare the Certificate Files

For Nginx, you need to concatenate your primary certificate and the intermediate certificates into a single file:

```
bash

cat your_domain_name.crt DigiCertHighAssuranceCA-3.crt >
/etc/ssl/certs/your_domain_name.bundle.crt
```

Ensure the certificates are in PEM format.

5. Configure Nginx to Use the SSL Certificate

Edit your Nginx server block configuration (usually found in `/etc/nginx/sites-available/your_site` or `/etc/nginx/conf.d/default.conf`):

```
nginx

server {
    listen 443 ssl;
    server_name your_domain_name;

    ssl_certificate /etc/ssl/certs/your_domain_name.bundle.crt;
    ssl_certificate_key /etc/ssl/private/your_domain_name.key;

    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_ciphers 'TLS_AES_128_GCM_SHA256:TLS_AES_256_GCM_SHA384:ECDHE-RSA-
AES128-GCM-SHA256:ECDHE-RSA-AES256-GCM-SHA384';

    location / {
        root /var/www/your_site;
        index index.html index.htm;
    }
}
```

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6. Test Nginx Configuration

Before restarting Nginx, test the configuration for syntax errors:

```
bash

sudo nginx -t
```

7. Restart Nginx

If the test is successful, restart Nginx to apply the changes:

```
bash

sudo systemctl restart nginx
```

8. Verify the SSL Installation

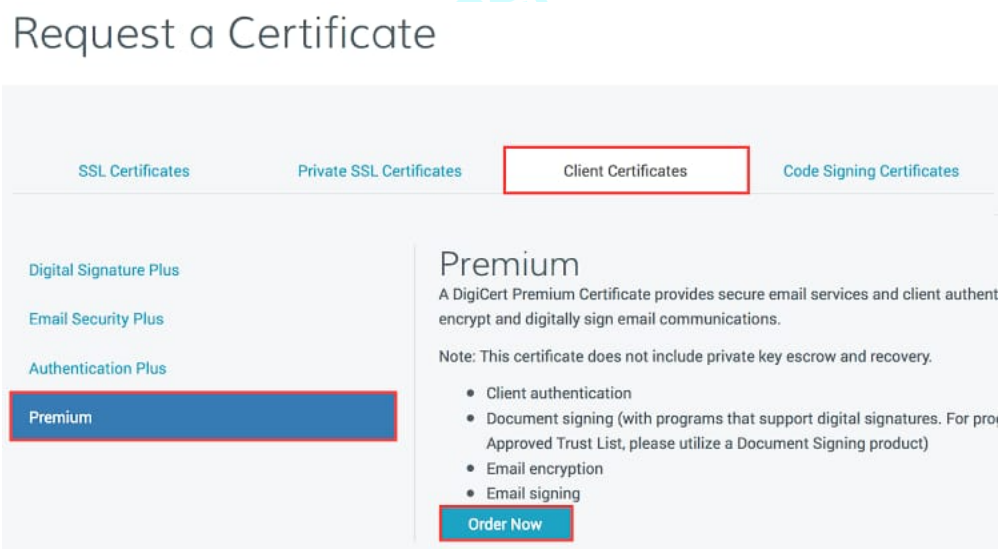
Use DigiCert's SSL Installation Checker to verify that your SSL certificate is correctly installed:

🔗 DigiCert SSL Installation Checker

✅ Summary

Step	Command/Action
1. Generate CSR	`openssl genrsa -out /etc/ssl/private/your_domain_name.key 2048` `openssl req -new -key /etc/ssl/private/your_domain_name.key -out /etc/ssl/private/your_domain_name.csr`
2. Submit CSR to DigiCert	Follow DigiCert's process to submit the CSR and complete domain validation
3. Download Certificate	Download `your_domain_name.crt` and intermediate certificates from DigiCert
4. Prepare Certificate Files	`cat your_domain_name.crt DigiCertHighAssuranceCA-3.crt > /etc/ssl/certs/your_domain_name.bundle.crt`
5. Configure Nginx	Edit Nginx config to point to the certificate and key files
6. Test Configuration	`sudo nginx -t`
7. Restart Nginx	`sudo systemctl restart nginx`
8. Verify Installation	Use DigiCert SSL Installation Checker

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DIGICERT Certificate Page

Creating and Installing a Self-Signed SSL Certificate (with Nginx)

🔍 What is a Self-Signed SSL Certificate?

A **self-signed SSL certificate** is an SSL certificate that is **signed by the creator rather than a trusted certificate authority (CA)**.

✅ Use Cases

- Development and local testing
- Internal applications (intranet)
- Prototyping SSL support

⚠️ Limitations

- Not trusted by browsers by default (causes warnings)
- Not recommended for public-facing websites

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🔧 Part 1: Creating a Self-Signed SSL Certificate

📌 Prerequisites:

- OpenSSL installed (default on most Linux/macOS; install manually on Windows)
- Terminal/command prompt access

📂 Step-by-Step:

1. Generate a Private Key

```
bash
openssl genrsa -out mysite.key 2048
```

2. Generate a Certificate Signing Request (CSR)

```
bash
openssl req -new -key mysite.key -out mysite.csr
```

Fill in details like:

- Common Name (use `localhost` or your dev domain)
- Country, State, Org, etc.

3. Generate the Self-Signed Certificate

```
bash

openssl x509 -req -days 365 -in mysite.csr -signkey mysite.key -out
mysite.crt
```

🎉 You now have:

- `mysite.key` – your private key
- `mysite.crt` – your certificate

🚀 Part 2: Installing SSL on an Nginx Server

📂 Folder Setup Suggestion

Move files to a secure directory:

```
Bash

sudo mkdir -p /etc/nginx/ssl/
sudo cp mysite.key mysite.crt /etc/nginx/ssl/
```

🧾 Configure Nginx

Edit your Nginx configuration (`/etc/nginx/sites-available/default` or your custom conf):

```
nginx

server {
    listen 443 ssl;
    server_name localhost;

    ssl_certificate /etc/nginx/ssl/mysite.crt;
    ssl_certificate_key /etc/nginx/ssl/mysite.key;

    location / {
        root /var/www/html;
        index index.html index.htm;
    }
}
```

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(Optional) Redirect HTTP to HTTPS:

```
Nginx

server {
    listen 80;
    server_name localhost;
    return 301 https://$host$request_uri;
}
```

🔁 Restart Nginx

```
bash
sudo nginx -t   # test config
sudo systemctl reload nginx
```

🧪 Part 3: Trust the Certificate in Browser

Chrome / Firefox (Linux/macOS/Windows):

1. Open browser settings → Privacy & Security → Certificates
2. Import `mysite.crt`
3. Mark it as **trusted for identifying websites**

macOS (Keychain Access):

- Open Keychain → Import `mysite.crt`
- Set to **Always Trust**

Windows:

- Run `certmgr.msc`
- Import the certificate into **Trusted Root Certification Authorities**

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✅ Summary

Task	Command
Generate key	`openssl genrsa -out mysite.key 2048`
Create CSR	`openssl req -new -key mysite.key -out mysite.csr`
Create certificate	`openssl x509 -req -days 365 -in mysite.csr -signkey mysite.key -out mysite.crt`
Configure Nginx	Point to `.crt` and `.key`
Trust browser	Import certificate manually